R-INSIGHT STUDIO

Building an Offline Data Analysis Studio

A portfolio case study on combining R, Shiny, data cleaning, statistics and visualization into one focused offline application.

R	Shiny	ggplot2	DT	readr / readxl	Git

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Project Type R Mini Project	**Interface** Shiny Application	**Data Model** Reactive Data Frame	**Launch** Local / Offline
Modules 10 Main Modules	**Analysis** Statistics / Correlation / Regression	**Visuals** 6 Chart Types	**Export** Clean CSV

PROJECT STORY

Why I Built R-Insight Studio

I wanted to build an R project that went beyond running isolated commands in the console. R-Insight Studio turns common data-analysis tasks into one connected interface where a user can import a dataset, inspect it, clean it, analyze it and export the result without repeatedly writing code.

The project became a practical exercise in reactive programming and modular design. Every screen works with the same current dataset, so cleaning or filtering data immediately affects statistics, graphs, correlation, regression and the final exported CSV.

CONTEXT

The Problem

Manual and fragmented analysis
Working directly in scripts or the console can make repeated importing, cleaning, plotting and statistical analysis harder for beginners to follow as one workflow.

What R-Insight Studio does
Brings dataset import, exploration, cleaning, operations, statistics, visualization and export into one local Shiny workflow using the same current dataset.

KEY CAPABILITIES

What the Application Can Do

Import & explore
Upload CSV or Excel files, preview records, inspect dimensions, missing values, duplicates and technical structure.

Cleaning & operations
Inspect problematic rows, replace missing values, remove duplicates, filter, sort, add, delete, rename and convert columns.

Statistics & visualization
Calculate descriptive statistics and create histograms, bar charts, box plots, scatter plots, line charts and pie charts.

Correlation & regression
Measure Pearson correlation, interpret strength and direction, and run simple linear regression with equation, R^2 and fitted line.

Hypothesis testing
Run a one-sample t-test with p-value, decision and a density graph showing critical regions and the test statistic.

Export & reset
Restore the original uploaded dataset when needed and export the latest cleaned or modified working dataset as CSV.

ARCHITECTURE

How R-Insight Studio Is Structured

R-Insight Studio follows a compact Shiny architecture: app.R connects the application, reusable Shiny modules handle each feature, reactive values hold the original and current datasets, and R packages provide importing, tables, statistics and visualization.

User UI → **Shiny Modules** → **Reactive Data** → **R Analysis** → **Updated View**

Layer	Main Technology	Responsibility
Interface	R Shiny	Tabs, inputs, buttons, tables, plots and user interaction.
Application	Shiny modules + reactive values	File upload, cleaning, operations, validation and shared dataset state.
Analysis & insight	Base R + ggplot2 + DT	Statistics, visualizations, correlation, regression, hypothesis testing and tables.

EXAMPLE WORKFLOW

Data Flow: Cleaning and Analyzing a Dataset

Upload file → **Create data frame** → **Clean / modify** → **Analyze** → **Visualize** → **Export CSV** → **Reuse result**

A dataset is uploaded once and stored as both an original copy and a current working copy. Cleaning and data-operation modules update the working copy, while statistics, charts, correlation, regression and hypothesis testing read from that same reactive dataset. The user can reset to the original data or export the latest version as CSV.

CORE RELATIONSHIPS

Shared Reactive Data Model

Entity	Connection	Usage
Original dataset	Uploaded file	Reset reference and data safety

Current dataset	Reactive working copy	Cleaning, operations and every analysis module
Numeric columns	Selected variables	Statistics, correlation, regression and t-test
Categorical columns	Selected groups	Bar and pie visualizations

LOCAL DATA ENGINEERING

The Data Work Behind the UI

R-Insight Studio deliberately avoids a database for its current scope. Uploaded files are loaded into R data frames, and reactive values keep the working dataset synchronized across modules. Small module functions keep importing, cleaning, analysis and presentation separated and easier to understand.

Object	Examples	Purpose
Imported data frames	CSV, XLS, XLSX	Load user data into R without requiring a database.
Reactive values	original_data, current_data	Share dataset state across modules and preserve a reset copy.
Module functions	Dashboard, cleaning, statistics, plots	Keep UI and server logic separated from the main app file.
Analysis helpers	mean, median, cor, lm, t.test	Turn selected columns into statistics, models and interpretable output.

RELIABILITY

Data Safety & Control

ORIGINAL COPY	REACTIVE STATE	VALIDATION	LOCAL EXPORT
Restore uploaded data	**Keep modules synchronized**	**Prevent invalid operations**	**Save cleaned CSV**

ENGINEERING CHOICES

A Few Design Decisions I'm Proud Of

Modular Shiny structure

Instead of putting the whole application inside one large app.R file, each major screen has its own UI and server module. This keeps the main file short and makes individual features easier to read, test and maintain.

One shared working dataset

The application keeps an original dataset and a current reactive dataset. Cleaning and data operations modify only the working copy, so every analysis module always uses the latest state while Reset can restore the original data.

Readable statistical output

Correlation, regression and hypothesis testing present useful interpretations and graphs rather than only raw console output. Technical R results remain available where they are useful for learning or demonstration.

Visual analysis

Charts use clear labels, markers, trend lines, value annotations and purpose-specific views so users can understand distributions, relationships and category proportions more easily.

SCOPE

What the Project Includes - and What It Does Not

Included	Intentionally out of scope
✓ CSV / Excel import and preview	– Database or user accounts
✓ Dashboard and data explorer	– Cloud deployment / live APIs
✓ Missing-value and duplicate cleaning	– Machine-learning models
✓ Filter, sort and column operations	– Advanced regression / testing
✓ Statistics and six chart types	– PDF or Word reports
✓ Correlation, regression and t-test	– Saved sessions between launches
✓ Clean CSV export	– Native Windows EXE packaging

REFLECTION

What I Learned

R-Insight Studio helped me understand how R can power an interactive application rather than only scripts and console analysis. The most useful lessons came from sharing reactive data between

modules, validating user operations and turning statistical output into visuals that are easier to interpret.

Shiny architecture Reusable modules, namespaces, reactive values and application flow.	Data cleaning Missing values, duplicates, filtering, sorting and datatype conversion.	Statistical analysis Descriptive statistics, Pearson correlation, regression and t-tests.
Data visualization Purpose-specific ggplot2 charts with labels, trends and annotations.	UX integration Readable summaries, inspection views, reset controls and validation messages.	Project delivery Git, GitHub, README, sample datasets and portfolio presentation.

CONCLUSION

Final Thoughts

R-Insight Studio began as a simple idea to import a dataset and analyze it, but it developed into a complete mini-project for exploring a practical R workflow. It combines file importing, data frames, cleaning, transformations, statistics, visualization and statistical testing in one local Shiny application.

For me, the strongest part of the project is the connected data flow: a change made during cleaning or filtering is immediately reflected in later statistics and graphs, and the same final dataset can be exported as CSV. That shared workflow makes the project feel like a small analysis studio rather than a collection of disconnected R examples.

EXPLORE

Project Links

GitHub	https://github.com/tushar-011
LinkedIn	https://www.linkedin.com/in/tushar-thakur-8848a7396/
Email	artificial.thakur@gmail.com